

Ouarezki A. Oussama

Data Scientist / AI / ML Researcher



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PROFILE

Master's student in Data Science at the University of Algiers 1, currently ranked first in a cohort of 90+ students. Specializing in RAG systems, with hands-on experience building neural reranking pipelines, interactive data dashboards, and AI-powered automation tools. Combines a strong academic foundation with practical skills in machine learning, NLP, and end-to-end system development.

MASTER'S THESIS

2025 – 2026
Algiers

- CERIST Research Center, Github** , *Reranking in Generative AI-Augmented Search (RAG)*
- Development of a containerized RAG web application (FastAPI), fully local to ensure data privacy, with a dashboard for monitoring retrieval and generation metrics.
 - Fine-tuning of a re-ranking model using LoRA and full domain adaptation, achieving performance close to proprietary solutions at a lower cost.
 - Design of an uncertainty-aware hybrid retrieval approach that reduces latency and cost while maintaining pipeline quality.
 - Extensible system (BI dashboard, domain-specific chatbot, or specialized RAG system).
 - Diagnostic tool for RAG pipelines, evaluating both generation quality and retrieval performance.

EDUCATION

06/2026
Algiers, Algeria

- Master's Degree in Data Science, University of Algiers 1**
- Expected to graduate 1st in cohort of 90+ students., Thesis co-supervised by Dr. Fatma Zohra Bessai-Mechmache (CERIST) and Dr. Khalida Guesmia (Univ. Algiers 1).

2024
Boumerdes, Algeria

Bachelor's Degree in Applied Mathematics, University of Boumerdes

TECHNICAL SKILLS

- Statistics & Math**
- Regression
 - Time series forecasting
 - Predictive modeling

- Libraries & Tools**
- HuggingFace
 - Git
 - PowerBI
 - Ollama

- AI / ML**
- RAG systems,
 - neural reranking
 - LLM fine-tuning (LoRA)
 - deep learning (PyTorch)
 - federated learning

- Programming**
- Python
 - HTML/CSS
 - Django

- Data Engineering**
- SQL, NoSQL
 - Apache Hadoop & Spark
 - Web Scraping
 - OCR / Document Parsing

- Deployment**
- Docker
 - FastAPI
 - Linux

LANGUAGES

Arabic

Native

English

Fluent

French

Intermediate

CERTIFICATION

Google Data Analyst [↗](#)

February 2025

KEY PROJECTS

Big Data Distributed Electricity Consumption (Time Series) - Using Spark and Hadoop [↗](#), [Report and video](#)

- Built an end-to-end distributed forecasting pipeline using Hadoop and Apache Spark and docker to simulate multiple machines.

Federated Learning for Enterprise Model Sharing [↗](#), [GitHub](#)

- Implemented a federated learning framework enabling multiple organisations to collaboratively train a shared model without exposing private data.

Text Extraction from Invoices [↗](#), [Report and Video](#)

- Automated structured data extraction from scanned invoices using OCR and regular expressions.

Social Network Recommendation System [↗](#), [Report and Video](#)

- Designed a graph-based recommendation pipeline implementing collaborative filtering and graph traversal algorithms to surface relevant connections, utilizing graph theory for analysis.

Data Science Foundations [↗](#), [Reports and Video](#)

- Comprehensive notebook series covering supervised and unsupervised learning, model evaluation, feature engineering, and visualisation -- designed as an accessible reference.

More Projects on GitHub [↗](#)

- Time Series Analysis and Prediction for CO2 Emissions / Multi-Agent Simulation in a Shoe Manufacturing Factory / Active Learning / Web Scraping and Sentiment Analysis / Algeria Telecom Dashboard Using Power BI / Exploratory Data Analysis for Student Performance / Data Analysis to Maximize Remote Job Opportunities / Data Warehouse for Cement Consumption and Power BI Dashboard / Text Classification / Active Learning / Semi-Supervised Learning / Optimization / COVID.NoSQL — Document-Oriented Database for Pedagogical Activity Tracking / Machine Learning From Scratch / Mathematical Visualization

SCIENCE COMMUNICATION

Stochastic Ways YouTube Channel [↗](#)

- Creator of the YouTube channel Stochastic Ways, dedicated to making mathematics, probability, and data science accessible to a broad audience. Produces original Manim-based mathematical animations that build intuition for complex concepts. Several projects are accompanied by full video explanations.
- Projects with video walkthroughs: Big Data & Spark forecasting, Text extraction from invoices, Social network recommendation, Data science foundations series.

Open-source tooling

- Author of manim-neovim - a Neovim plugin for running and previewing Manim animations directly from the editor. Also maintains a custom Neovim config optimised for data science and Typst / LaTeX workflows.
- manim-neovim on GitHub